

Fluid Dynamics of the Atmosphere and Ocean

Problem Set 2

Unassessed.

1. In a two-layer shallow water system in geostrophic balance, show that the difference in the velocity between the two layers is proportional to the slope of the interface between the layers.
2. (a) Model the atmosphere as two immiscible, 'shallow water' fluids of different density stacked one above the other. Suppose that the vertical shear of the horizontal wind (i.e., the difference between the zonal wind in the two layers) is 20 m s^{-1} . Using reasonable values for any required physical parameters, estimate the slope of the interfacial surface, and hence how much the surface rises between equator and pole.
(b) Similarly estimate an interfacial displacement in the ocean associated with a velocity shear of 1 m s^{-1} over a distance of 4000 km. (This is a crude representation of the main oceanic thermocline.)
3. (a) Briefly discuss how the shallow water equations are similar or dissimilar to the compressible fluid equations with an equation of state of the form $p = f(\rho)$ or (if you wish) $p = A\rho^\gamma$, where A and ρ are constants.
(b) Show that the linearized shallow water equations are *isomorphic* to the linearized compressible fluid equations for a particular value of γ , and state that value.
4. Show that the vertical velocity within a shallow water system is given by

$$w = \frac{z - \eta_b}{h} \frac{Dh}{Dt} + \frac{D\eta_b}{Dt}. \quad (1)$$

Interpret this result, showing that it gives sensible answers at the top and bottom of the fluid layer. If you wish, simply do the problem with $\eta_b = 0$.